

Thesis figures for draft3: what is in, what is out, and where the rest goes

Generated 31 July 2026 from `draft3.md` (1607 lines), the `Doc images/` and `Other images/` folders, `HANDOFF_MORNING_dome_photo.md`, and `THESIS_REVISION_PLAN.md`. This replaces the 30 July guide, which was written against `thesis1.md` and is now partly out of date. Numbers quoted in captions were measured by running the code, not estimated.

What changed since the last guide. Four things, and they matter for how you read the rest of this document:

- **The dome photographs exist now.** The old guide opened with "three figure slots need photographs of the built dome that do not exist". `draft3` has Figures 4.1, 4.2, 4.3 and 4.5, all photographic. **That gap is closed.**
- **Chapter 4 grew from four figure slots to five**, and the numbering collides with itself in three places. See Section D.
- **The generated figures are still almost all uninserted.** Eleven of the thirteen `NEW_fig_*` files in `Doc images/` have no slot in `draft3` yet.
- **Section 6.4 still does not exist**, and §7.1 still points at it. The photograph you take in the morning goes there. See Section C.

Read first. Three of the photographs now in `draft3` are *not in this project folder*. Neither `Doc images/` nor `Other images/` contains any picture of an assembled dome; the July photographs there are the film delivery, the LCD axis tests and the sky axis tests. So Figures 4.1, 4.2 and 4.5 came from somewhere else, probably straight off your phone into Overleaf. **Copy them into `Doc images/`** so they are backed up with the rest of the thesis and so the next pass can check them.

A. The eleven slots draft3 already has, and what should be in each

Fig	Caption in draft3	File that belongs in the slot	Check
3.1	Solar noon elevation across 2026 at Turku	outputs/sun_position_validation.png	OK Caption claims agreement within 0.01°. True for 52.986; the draft's other number, 52.97 at line 616, is stale. Quote 6.110 and 52.986.
3.2	Geometry of the sky polarization pattern, panels (a) and (b)	Doc images/fig_sky_polarization_geometry.png	OK Caption is correct and contradicts the body text at line 731. Fix the text, not the caption.
4.1	Construction sequence of the dome	Photograph. Not in the project folder.	COPY IN Also see Section D: this should probably not be 4.1.
4.2	Image of final version of the constructed dome	Photograph. Not in the project folder.	DUPLICATE Same subject as 4.5. Keep one.
4.3	Testing the polarizing film and confirming the polarity	Doc images/WhatsApp Image 2026-07-29 at 12.37.23.jpeg	OK Caption undersells it. Rewrite per Section F.
4.4	Gore template, sheet layout, and assembled hemisphere from the dome cutting guide	Doc images/NEW_fig_4_1_gore_template_R40.png	VERIFY The caption says "assembled hemisphere", which the generated figure does not show. If you pasted the diagram out of dome_cutting_guide_FINAL_CONFIRMED.pdf instead, swap it: only the generated file carries the R = 40 mm numbers on the drawing.
4.5	Final Dome Construction Prototype	Photograph. Not in the project folder.	DUPLICATE See 4.2.
5.1	Bench test capture through the bare fisheye lens	captures/br_50/sky_20260726T152018Z.png	KEEP Keep the "optic not fitted" caption. It becomes the control half of the pair once the dome frame exists.
6.1	Daylight solar positions at Turku, two panels	outputs/turku_sun_coverage.png	OK Caption says "35 to 325 degrees"; the estimator set is 50 to 311. Both populations, one sentence. Section F.
6.2	Modelled sky radiance through the polarising optic	outputs/polarizer_band_demo.png	RE-INSERT The version in your draft has the wrong claim printed inside the picture. The regenerated file is below.
6.3	Synthetic frames rendered through the flat filter	outputs/matched_synthetic_example.png	CAPTION "Because ordinary photographs record no polarisation information" needs the qualifier. Section F.

Figures 4.2 and 4.5 are the same picture twice. "Image of final version of the constructed dome" and "Final Dome Construction Prototype" are two captions for one subject, on consecutive pages, and the body text at line 890 says "figure 4.2 shows the final prototype", which points at the first while describing the second. Delete one slot, or make them genuinely different: one looking down the optical axis so the eight gores and the apex are visible, one from outside in daylight on the mount.

B. The eleven figures still waiting in [Doc images/](#)

File	Goes in	Why
NEW_fig_3_1_azimuth_conventions.png	§3.1, after line 646	The South-from-West vs clockwise-from-North confusion, which recurs in four chapters.
NEW_fig_3_3_dop_vs_scattering_angle.png	§3.3, after the DoP equation (line 768)	The equation currently stands alone on the page.
NEW_fig_4_4_processing_pipeline.png	§4.4, fills the placeholder at line 949	The only placeholder left in Chapter 4.
NEW_fig_5_1_exposure_bracket.png	§5.1, with the exposure discussion (line 1047)	Turns the exposure argument into measured data.
NEW_fig_6_2_azimuth_profile.png	§6.2, after Figure 6.2	The measurement that settles where the dark lobe actually falls.
outputs/polarizer_band_demo.png	§6.2, replaces Figure 6.2	Corrected caption, and the solar meridian is now drawn on it.
NEW_fig_6_3a_error_by_condition.png	§6.3, before Table 6.1	Visual summary of the six-condition matrix.
NEW_fig_6_3b_error_distributions.png	§6.3, after Table 6.1	Shows the bimodality that explains the mean/median gap.
NEW_fig_6_3c_error_vs_elevation.png	§6.3, optional	Ties accuracy to the high-latitude elevation range. First to drop if pages run short.
NEW_fig_7_1_confidence_vs_error.png	§7.1	Evidence for the confidence-metric limitation you now admit in the conclusion.
NEW_fig_7_2_seasonal_solar_arcs.png	§7.2, with the October note (line 1354)	Fills one of the two remaining NOTE blocks in Chapter 7.

NEW_fig_6_4a_exposure_disc_tradeoff.png and NEW_fig_6_4b_captured_frames.png are deliberately absent from this table. They describe the 31 bare-lens frames and were written for a §6.4 that had no dome photograph. Section C explains what happens to them.

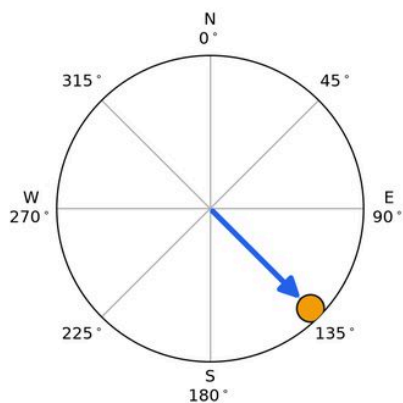
§3.1, after line 646, immediately after the paragraph on conventions

Caption: The two azimuth conventions used in this thesis, shown for a single physical sun direction. The derivation of Section 3.1 measures azimuth from South toward West, while every heading reported from Chapter 4 onward is measured clockwise from North. The two differ by a fixed rotation of 180 degrees.

The two azimuth conventions in use, shown for one physical sun direction. They differ by exactly 180 degrees, with the North referenced azimuth equal to the South referenced azimuth plus 180 reduced into 0 to 360. A mixed convention still runs and still returns a plausible number.

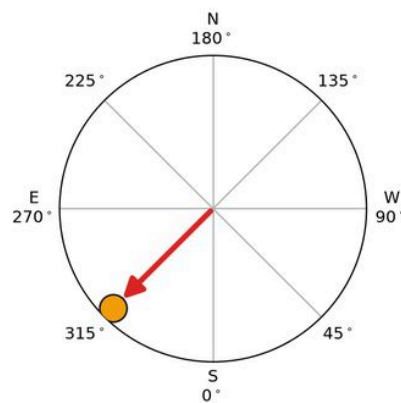
Chapters 4 to 7: robot convention
azimuth clockwise from North

the same sun reads 135 degrees



Section 3.1: supervisor's formulation
azimuth from South toward West

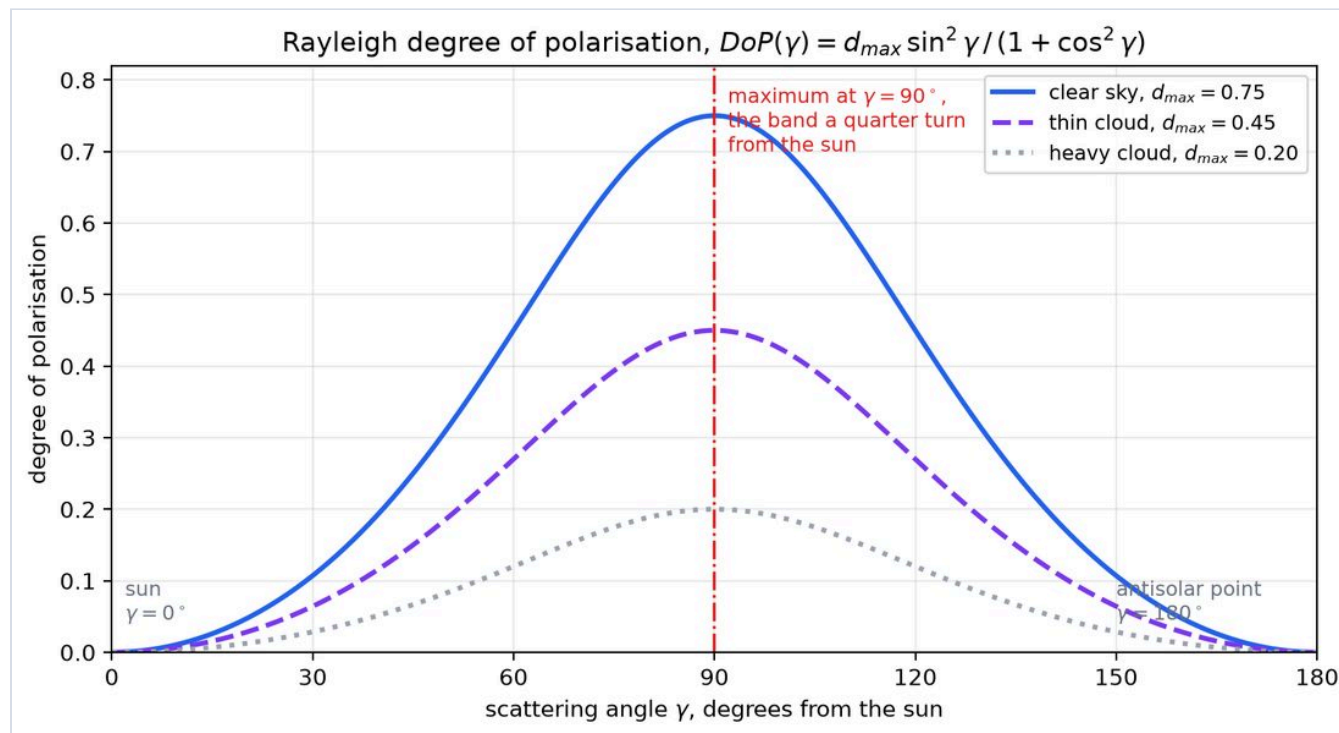
the same sun reads 315 degrees



NEW_fig_3_1_azimuth_conventions.png

§3.3, immediately after the equation for DoP(gamma) at line 768

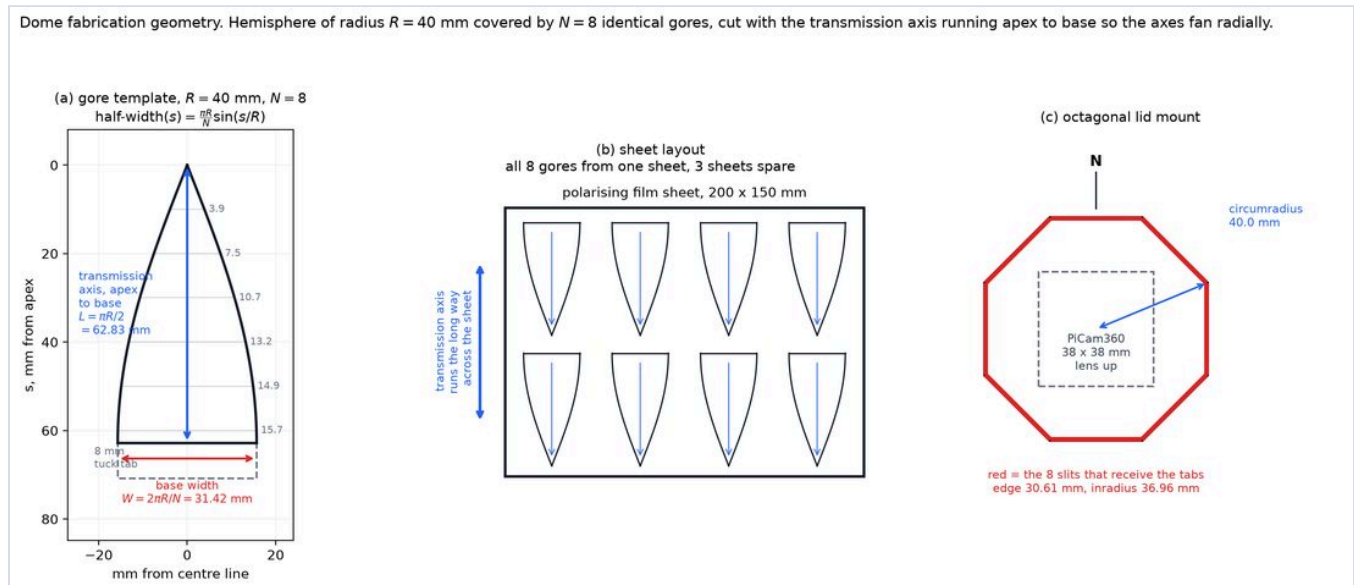
Caption: The Rayleigh degree of polarisation as a function of scattering angle, for the three values of the maximum degree of polarisation used in the simulation experiments of Chapter 6. The degree of polarisation vanishes toward the sun and the antisolar point and peaks along the band lying a quarter turn from the sun.



NEW_fig_3_3_dop_vs_scattering_angle.png

§4.2, the figure that should carry the $R = 40$ mm geometry

Caption: Fabrication geometry of the polarising dome. The hemisphere of radius 40 mm is covered by eight identical gores, each of length 62.83 mm and base width 31.42 mm, whose half-width at a distance s from the apex follows $(\pi R / N) \sin(s / R)$. All eight gores are cut from a single 200 by 150 mm sheet with the transmission axis running from apex to base, so that in the assembled dome the axes fan radially outward. The base tabs fold through eight slits cut along the edges of a regular octagon of circumradius 40 mm.

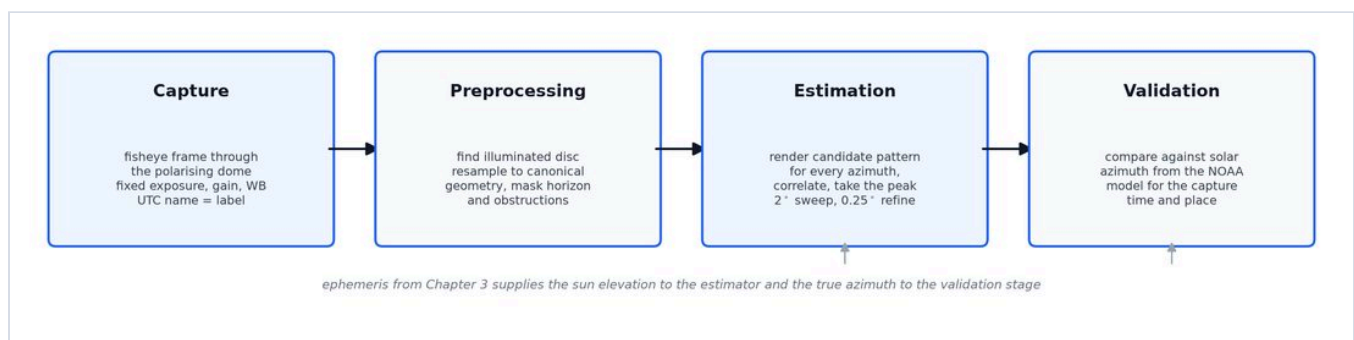


NEW_fig_4_1_gore_template_R40.png

Compare this against whatever is currently in the Figure 4.4 slot. If your version came from the cutting-guide HTML it will not have the measured numbers printed on it, and §4.2 is the one place in the thesis where those numbers have to be visible.

§4.4, fills the placeholder at line 949

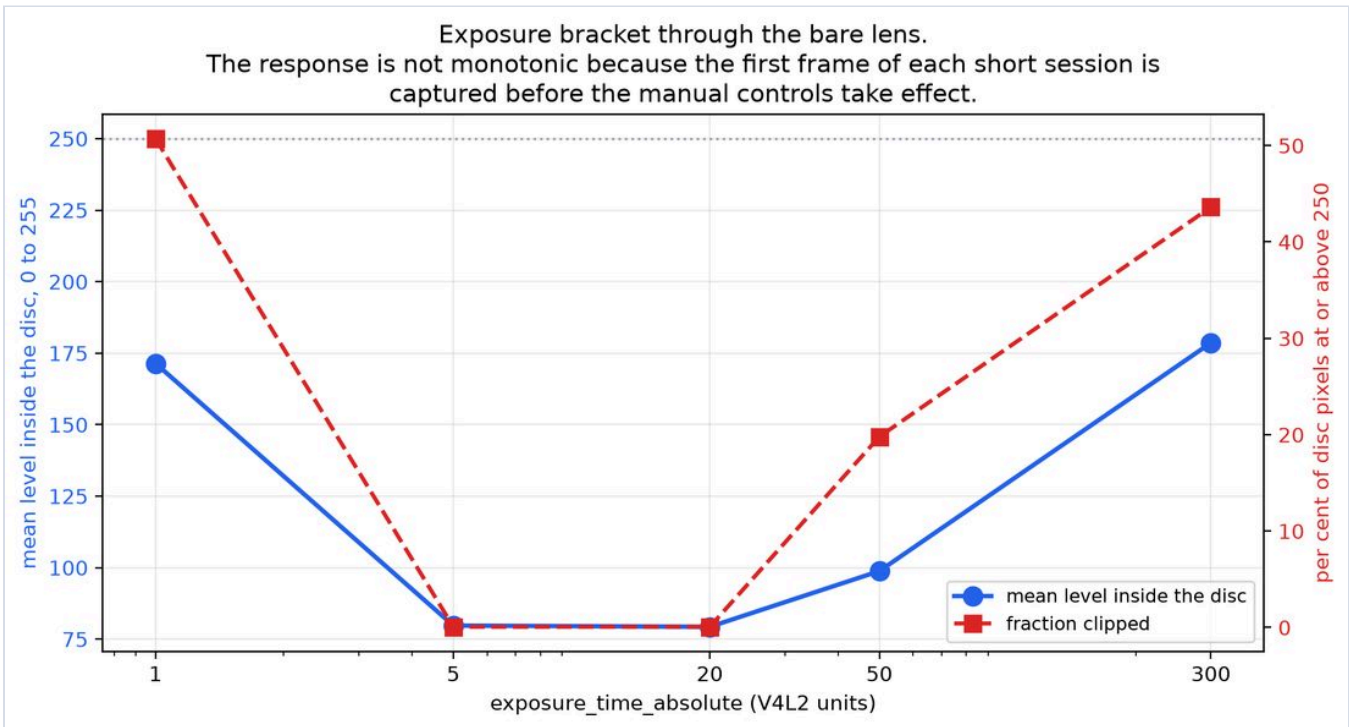
Caption: The processing pipeline. A frame is captured through the dome at fixed exposure and white balance and named with its coordinated universal time, the sky region is isolated and each retained pixel associated with a sky direction, the observed pattern is compared against a family of rendered patterns to yield an azimuth and a confidence, and the result is validated against the solar azimuth computed for the time and place of capture.



NEW_fig_4_4_processing_pipeline.png

§5.1, with the exposure discussion around line 1047

Caption: Exposure bracket through the bare fisheye lens, measured inside the fitted disc. The response is not monotonic, because the first frame of each short session is captured before the manual controls take effect, which is the settling behaviour described in the text.

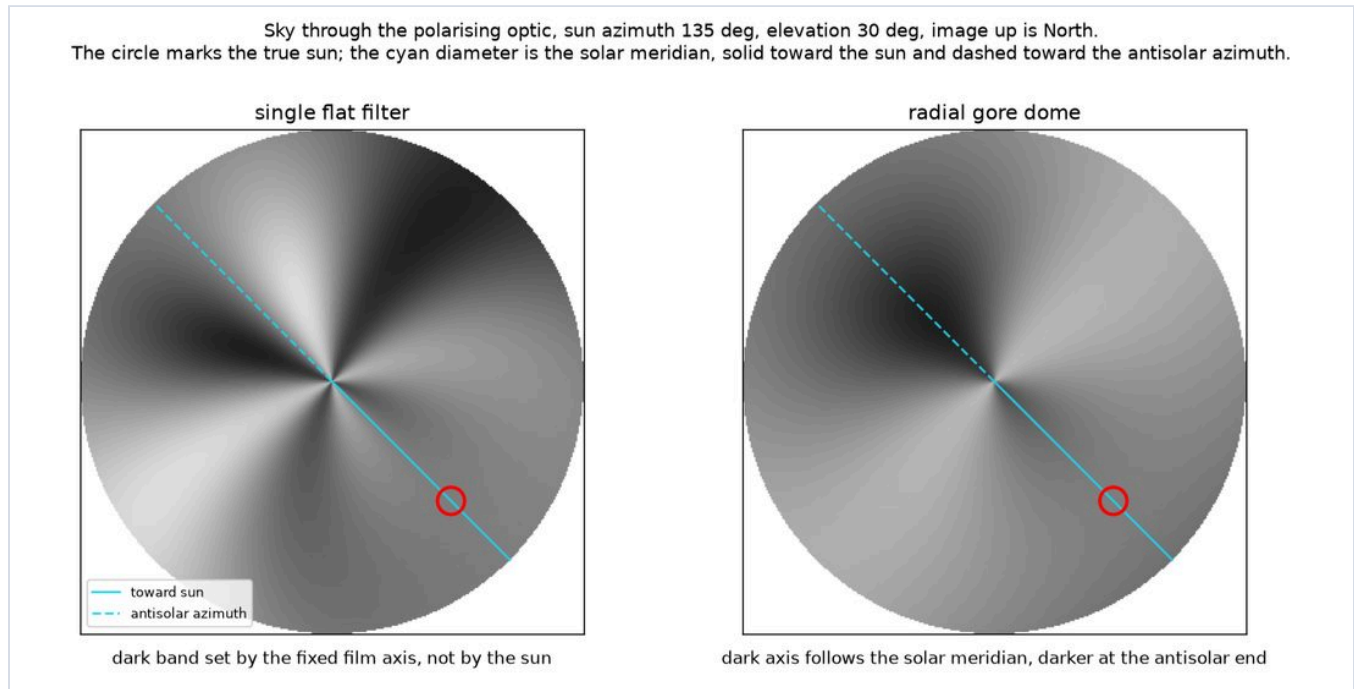


NEW_fig_5_1_exposure_bracket.png

Two numbers in §5.1 do not reproduce. Line 1040 says the sky region of Figure 5.1 has "an average intensity of 220 and about 1/3 of the sky pixels being 250 or higher". Measured inside the fitted disc of that frame: mean **158.5**, **27.6 per cent** at or above 250. The one-third figure is close; the mean is not. Line 1060 says "the difference in intensity between frames taken 3 seconds apart was about 2.6" — say "a factor of about 2.6" and quote the two means; the clearest instance in your data is the 26 July pair, where the mean fell from 152.0 to 68.3 across two seconds, a factor of 2.2.

§6.2, replaces the existing Figure 6.2

Caption: Modelled sky radiance through the polarising optic for a sun azimuth of 135 degrees and an elevation of 30 degrees, with image up corresponding to North. The left panel is the single flat filter and the right panel is the radial gore arrangement. The red circle marks the true solar position and the cyan diameter is the solar meridian, solid toward the sun and dashed toward the antisolar azimuth. Note that the antisolar *point* lies below the horizon and is not in the image; what is marked is the antisolar azimuth at the disc edge.

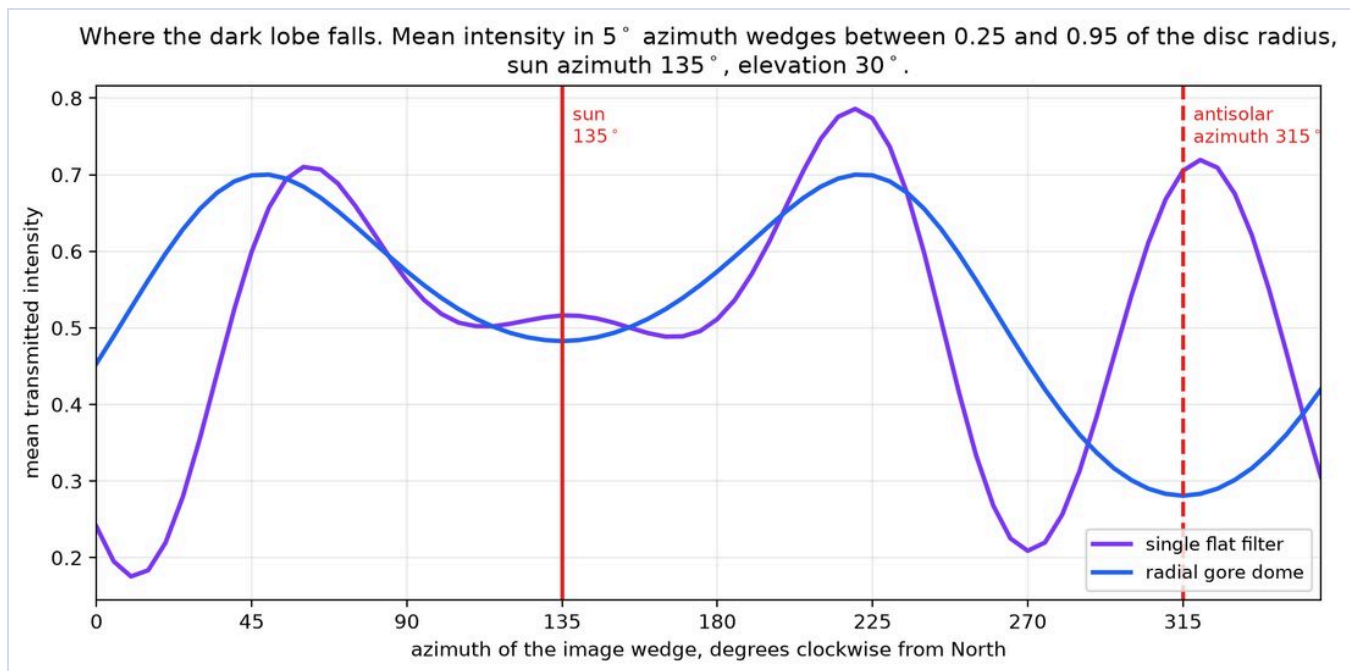


outputs/polarizer_band_demo.png

Why re-inserting this matters. The copy in your draft has the old claim printed *inside* the image, where an examiner reads it straight off the figure rather than off your prose. The regenerated file above says the correct thing and draws the meridian, so the claim is visible rather than merely asserted.

§6.2, after the replaced Figure 6.2

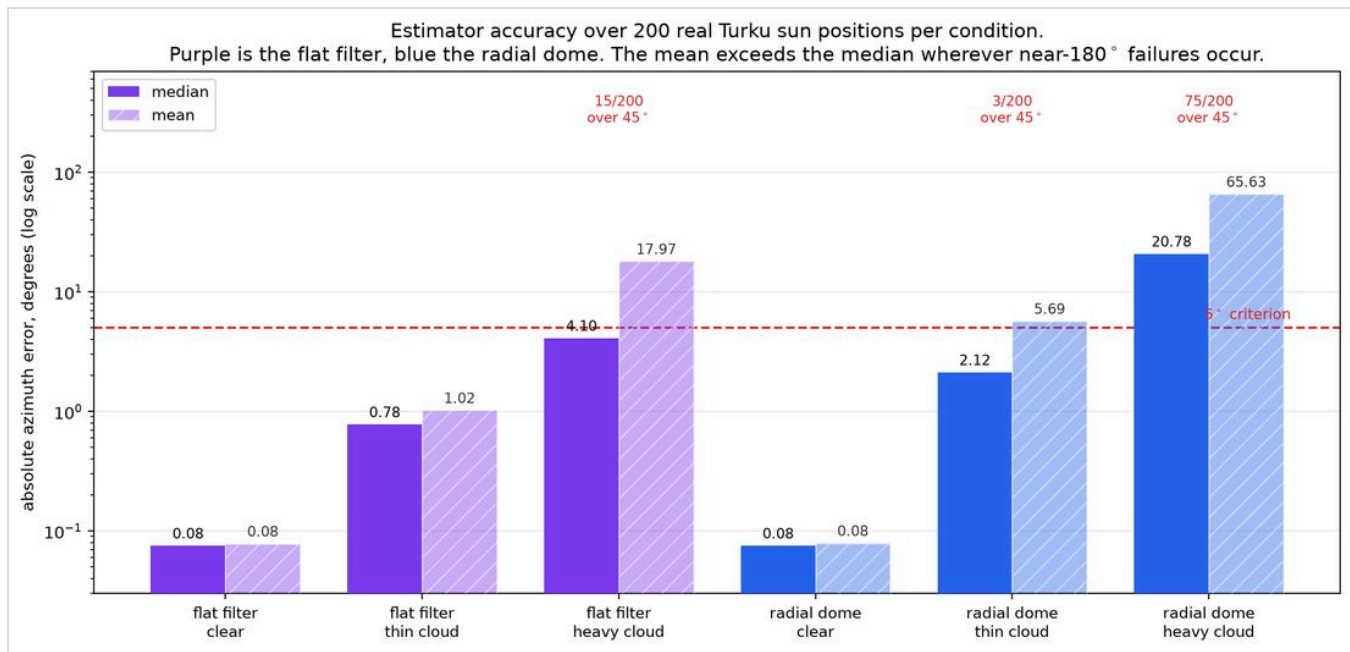
Caption: Mean transmitted intensity in five degree azimuth wedges between 0.25 and 0.95 of the disc radius, for a sun at azimuth 135 degrees and elevation 30 degrees. The flat filter produces two minima fixed by the film axis and unrelated to the sun. The radial dome produces a single deep minimum at the antisolar azimuth of 315 degrees together with a shallower one toward the sun, so the dark axis marks the solar meridian while the asymmetry between its ends resolves which end holds the sun.



NEW_fig_6_2_azimuth_profile.png

§6.3, before Table 6.1

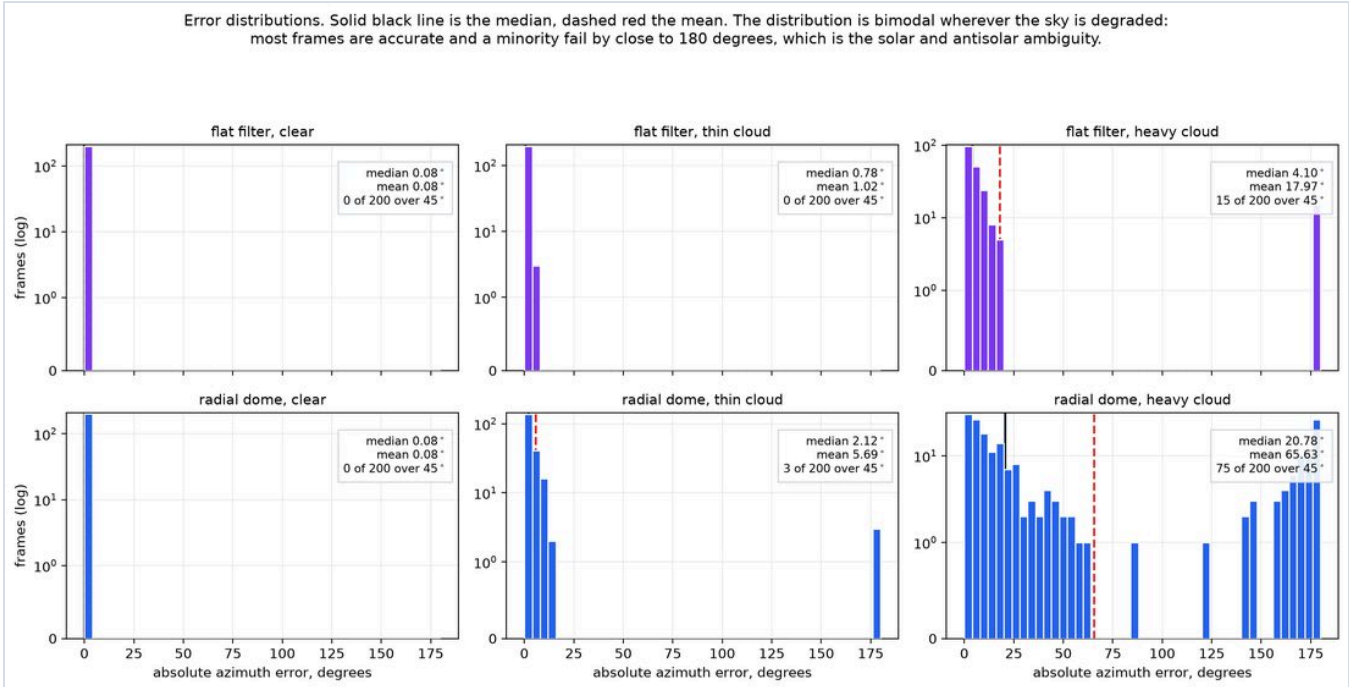
Caption: Absolute azimuth error over 200 sun positions drawn from the real Turku daylight record, for the two optical arrangements under three sky conditions. Bars show the median and the mean on a logarithmic scale. The mean exceeds the median wherever near-180 degree failures occur, and the count of such failures is given above each affected condition.



NEW_fig_6_3a_error_by_condition.png

§6.3, after Table 6.1

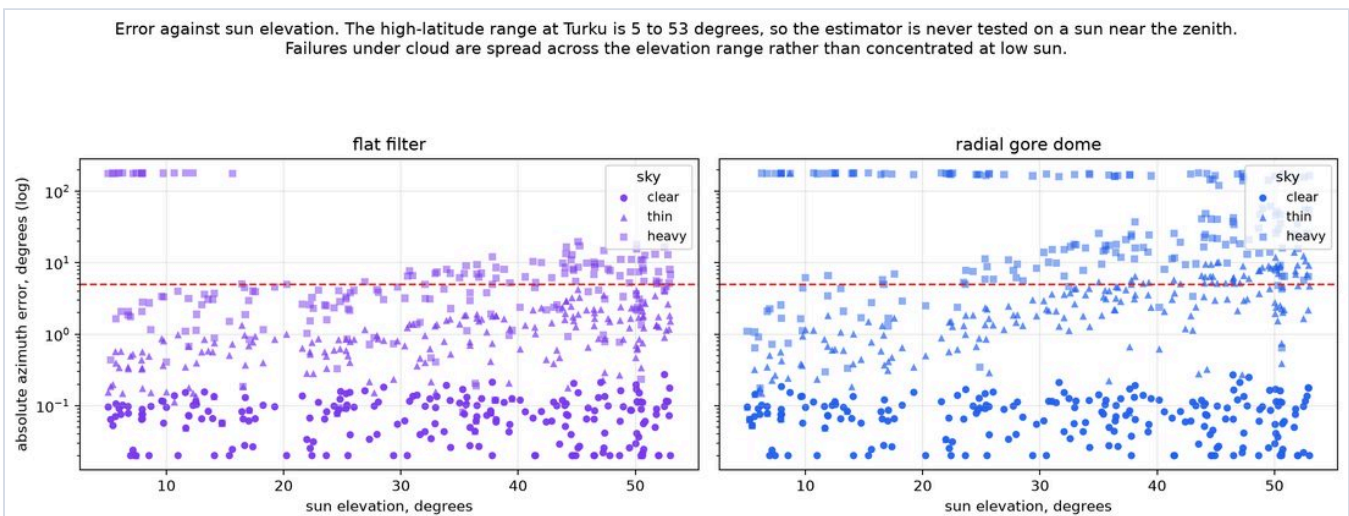
Caption: Distribution of the absolute azimuth error in each condition. The vertical scale is logarithmic. Under a clear sky the error is tightly concentrated below a tenth of a degree, while under cloud the distribution becomes bimodal, with most frames accurate and a minority failing by close to 180 degrees, which is the solar and antisolar ambiguity discussed in Section 7.1.



NEW_fig_6_3b_error_distributions.png

§6.3, optional

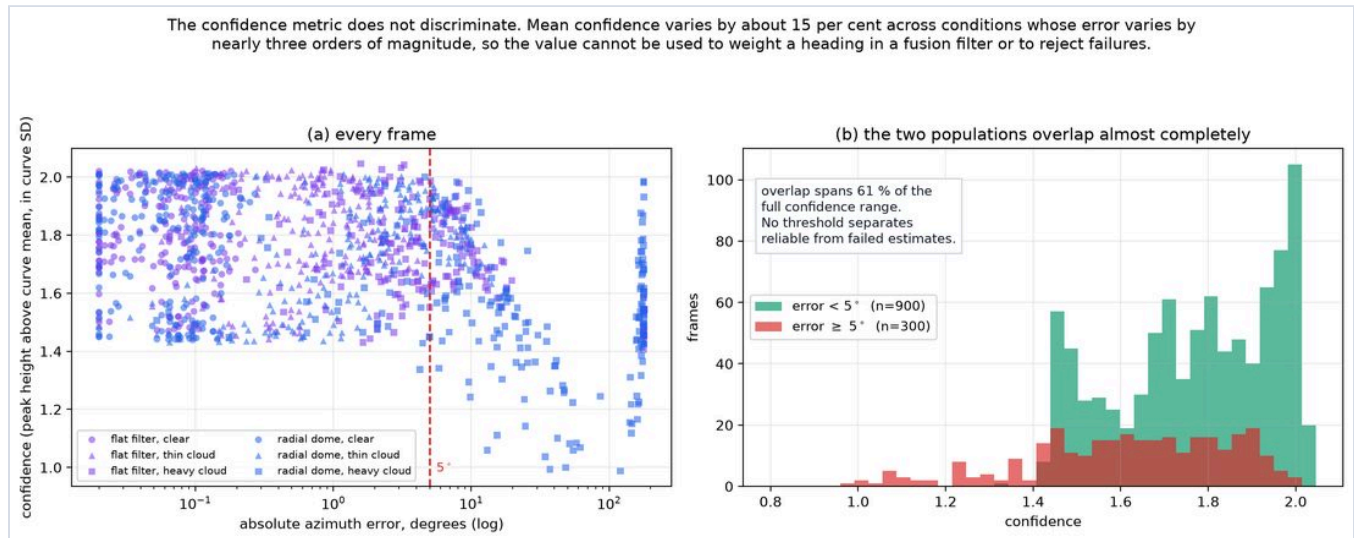
Caption: Absolute azimuth error against sun elevation. At Turku the elevation never exceeds 53 degrees, so the estimator is never tested against a sun near the zenith. Failures under cloud are spread across the elevation range rather than concentrated at low sun.



NEW_fig_6_3c_error_vs_elevation.png

§7.1, with the confidence-metric limitation

Caption: The confidence metric fails to separate reliable estimates from failed ones. Across conditions whose mean error spans nearly three orders of magnitude the mean confidence varies by about fifteen per cent, and the distributions of confidence for accurate and for failed frames overlap across most of the range. Since the purpose of the value was to weight a heading within a fusion filter and to reject solar and antisolar failures, neither is possible until the metric is redefined.



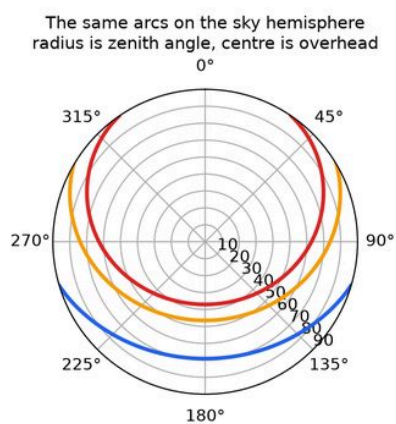
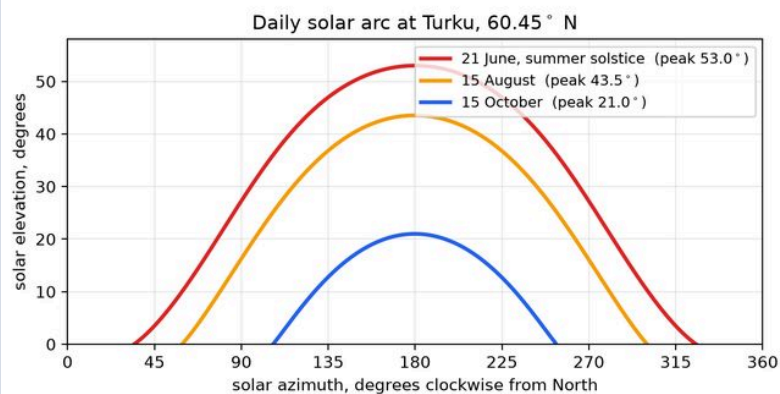
NEW_fig_7_1_confidence_vs_error.png

Numbers for the §7.1 text. Mean confidence by condition: 1.80, 1.80, 1.75, 1.78, 1.74, 1.53, against mean errors from 0.078 to 65.6 degrees. Per frame, estimates within 5 degrees have confidence 1.773 ± 0.179 and those beyond 5 degrees 1.620 ± 0.227 , so the two populations overlap across 61 per cent of the range. **Route forward to offer:** the ratio of the primary peak to the second highest local maximum would measure the solar and antisolar ambiguity directly, which is the failure that actually occurs, whereas peak height above the mean does not.

§7.2, fills the NOTE block at line 1354

Caption: Seasonal change in the sun geometry at Turku. In October the sun rises later and further south, reaches a noon elevation near 20 degrees rather than 53, and sweeps a much narrower range of azimuths, which moves the polarisation pattern toward the horizon where the analysis mask discards it. Repeating the experiments then would test the method against a second and quite different sun geometry at the same site.

Seasonal change in the sun geometry the compass must read. In October the sun rises later and further south, peaks near 20 degrees rather than 53, and sweeps a much narrower azimuth range, which pushes the polarisation pattern toward the horizon.



NEW_fig_7_2_seasonal_solar_arcs.png

C. The dome photograph, and what it does to Section 6.4

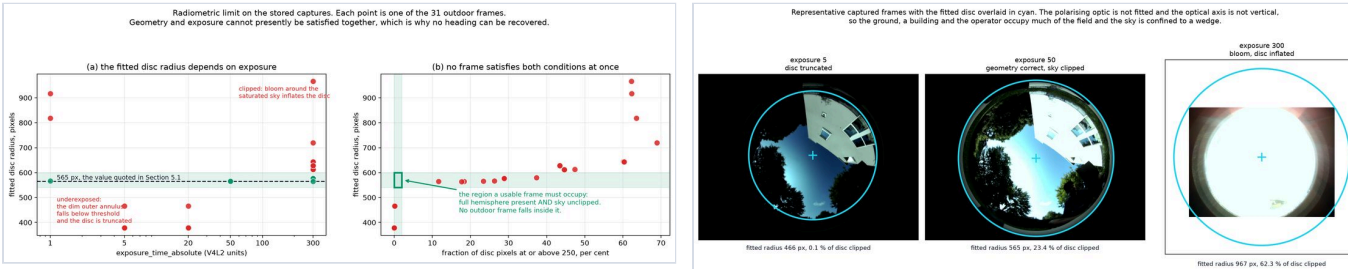
§7.1 line 1310 says "only the captured image validation of Section 6.4 tests the latter", and Section 6.4 does not exist in draft3. Chapter 6 stops at 6.3. This is a broken cross-reference an examiner will hit, and it is the single largest hole in the figure set.

The photograph you take through the dome goes into a new **§6.4, Captured image validation**, as **Figure 6.4**, and it should be the **two-panel control pair**: one frame through the dome and one bare-lens frame of the same sky, seconds apart. The pair, not the dome frame alone, is what answers your supervisor's question, because it shows the pattern appearing and disappearing with the optic and nothing else changed. A single dome frame is much weaker, since a reader can attribute the structure to sky brightness gradients.

Caption to write once you have it: The same sky recorded seconds apart without and with the polarising dome, at [time] UTC on [date] at Turku, with the camera heading set to [heading] degrees from North by compass and the lid levelled. The bare-lens frame records intensity alone. The frame through the dome shows the transmitted intensity pattern of Section 3.3, with the dark axis lying along the solar meridian. The two frames differ only in the presence of the optic, so the pattern in the second is produced by the polarisation of the skylight and not by the brightness of the sky.

Where the two old §6.4 figures go

File	If the dome frame works	If the sky is overcast and you get nothing
NEW_fig_6_4a_exposure_disc_tradeoff.png	Move to §5.1 next to the exposure bracket, as the radiometric argument for why exposure is a calibration step. It is still true and still useful there.	Keep in §6.4 as the central result, exactly as the old guide had it.
NEW_fig_6_4b_captured_frames.png	Drop it. Figure 6.4 supersedes it, and three bare-lens frames with a building and the operator in shot will weaken a section that now has a real measurement.	Keep in §6.4.



Left: `NEW_fig_6_4a_exposure_disc_tradeoff.png` Right: `NEW_fig_6_4b_captured_frames.png`

Three consequential text edits once the photograph exists, all short: §7.1 first limitation at line 1305; the Conclusion at line 1400, which currently claims only that a prototype was "tested in camera to establish the image-acquisition chain"; and one sentence in §5.1 after line 989 naming Figure 5.1 as the bare-lens control. The abstract needs nothing: draft3 already says the results "provide a foundation for further outdoor validation", which survives.

D. Numbering, and a clean set

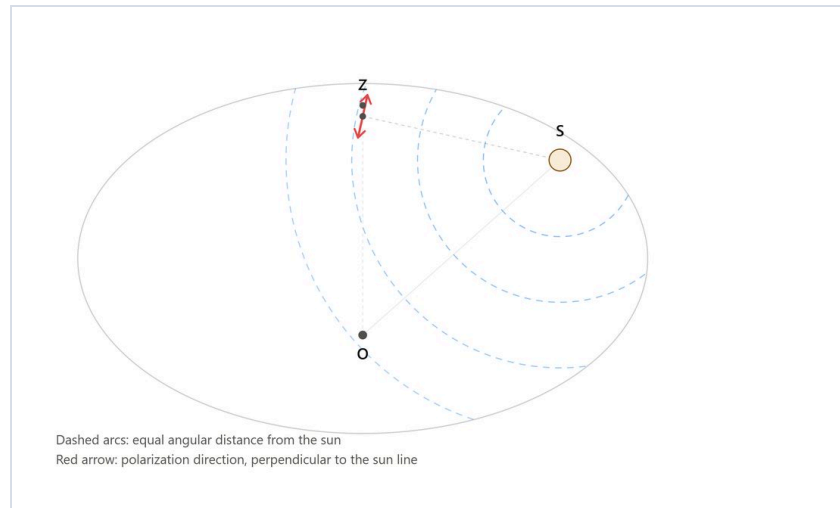
Chapter 4 numbering collides three ways in draft3. Line 890 says "figure 4.2 shows the final prototype", but Figure 4.2 is captioned "Image of final version of the constructed dome" and Figure 4.5 is "Final Dome Construction Prototype". The dead placeholder at line 895 calls the gore template "Figure 4.1" when it sits in the Figure 4.4 slot. The placeholder at line 949 calls the pipeline "Figure 4.2". Renumber once, at the end, after every figure is placed.

New	Figure	Source
3.1	Solar noon elevation across 2026	outputs/sun_position_validation.png
3.2	The two azimuth conventions NEW	NEW_fig_3_1_azimuth_conventions.png
3.3	Geometry of the sky polarization pattern	fig_sky_polarization_geometry.png
3.4	Degree of polarisation vs scattering angle NEW	NEW_fig_3_3_dop_vs_scattering_angle.png
4.1	Gore template, sheet layout, octagonal lid mount	NEW_fig_4_1_gore_template_R40.png
4.2	Construction sequence of the dome	photograph, currently 4.1
4.3	The assembled dome on its mount	photograph, merge of 4.2 and 4.5
4.4	Film transmission axis confirmed against the sky	WhatsApp ... 12.37.23.jpeg
4.5	Processing pipeline NEW	NEW_fig_4_4_processing_pipeline.png
5.1	Bench test through the bare fisheye lens	captures/br_50/...
5.2	Exposure bracket NEW	NEW_fig_5_1_exposure_bracket.png
6.1	Daylight solar positions at Turku	outputs/turku_sun_coverage.png
6.2	Modelled sky radiance through both optics	outputs/polarizer_band_demo.png
6.3	Azimuth wedge profile NEW	NEW_fig_6_2_azimuth_profile.png
6.4	Synthetic frames at true solar positions	outputs/matched_synthetic_example.png
6.5	Error by condition NEW	NEW_fig_6_3a_error_by_condition.png
6.6	Error distributions NEW	NEW_fig_6_3b_error_distributions.png
6.7	Error against sun elevation, optional NEW	NEW_fig_6_3c_error_vs_elevation.png
6.8	Control pair, sky without and with the dome NEW	the photograph you take
7.1	Confidence against error NEW	NEW_fig_7_1_confidence_vs_error.png
7.2	Seasonal solar arcs NEW	NEW_fig_7_2_seasonal_solar_arcs.png

That is 21 figures for a 49 page thesis, which is on the high side but not unusual for an instrument-building thesis. If you need to cut, drop 6.7 first, then 3.4.

Unused files in Doc images/

`sky_dome_iso_distance_arcs_and_aop.png` has no slot in the plan above. It overlaps Figure 3.3 panel (b). Leave it out unless you decide it explains the iso-distance arcs better than the panel you have; do not use both.



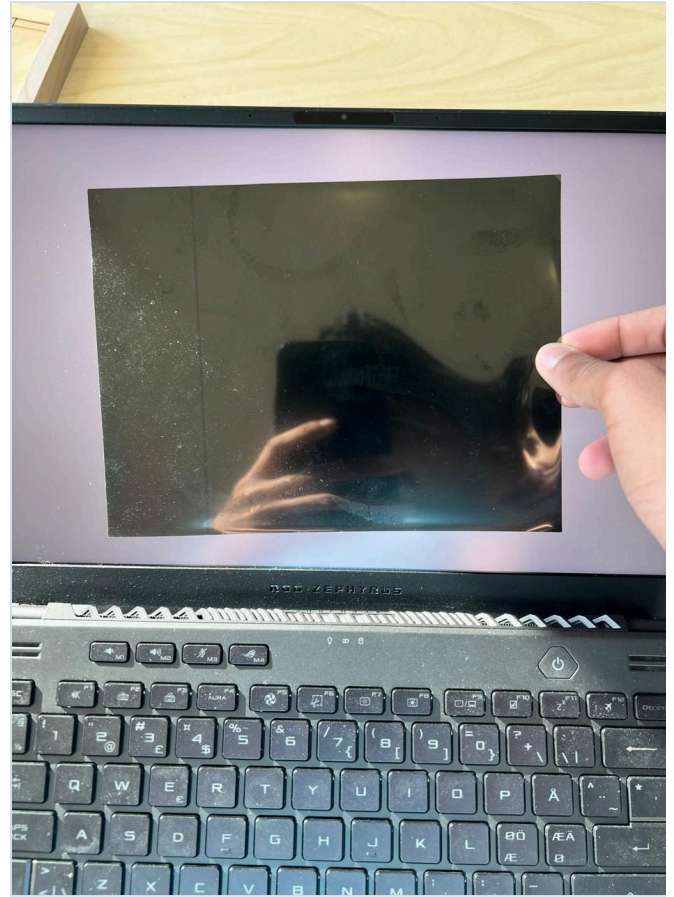
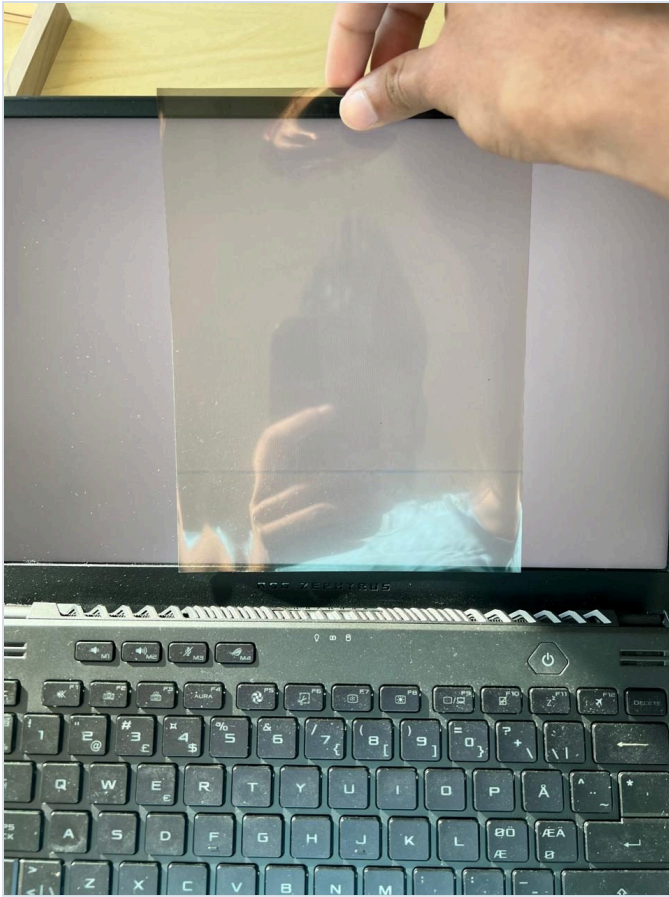
sky_dome_iso_distance_arcs_and_aop.png

E. The photographs you already have, for the record

These are the axis-determination photographs in the project folder. Together they are the three independent observations that resolve the 90 degree ambiguity a crossed-polariser test leaves, and they are much stronger evidence than the single sentence §4.2 currently gives them.



WhatsApp Image 2026-07-29 at 12.37.23.jpeg and ... 12.37.48.jpeg — two sheets against clear blue sky, one transmitting and one blocking, with the blocking sheet arrowed. The un-annotated higher resolution version of the same test is Other images/WhatsApp Image 2026-07-28 at 20.46.01.jpeg , which is the better one to print.



WhatsApp Image 2026-07-29 at 19.36.46.jpeg and ... (2).jpeg — the same sheet over an LCD panel whose polarisation axis is known, passing in one orientation and extinguishing in the other. This is the observation that fixes which of the two candidate axes is the pass axis.

Replacement for the axis paragraph in §4.2, line 899. The draft says the axis was found "with a simple crossed polarizer test". That test leaves a 90 degree ambiguity between the pass axis and the block axis, and you resolved it with three independent observations: glare off a horizontal surface, an LCD panel of known axis, and polarising sunglasses. All three agreed that the axis runs parallel to the long 200 mm edge of the sheet. Write that up; it is a much stronger methods paragraph and you have the photographs for it.

F. Text that has to change with the figures

Placeholders still live in draft3

Line	Where	What it is
492	Table 2.1	Your own row still reads "Chapter??" . Fill it, and note that every other row reports a <i>field-measured</i> accuracy while yours is <i>simulation</i> .
891–897	§4.2	NOTE block plus a dead figure placeholder. The photographs it asks for now exist, so delete the whole block once the measured radius is written into the text.
949–950	§4.4	"[Figure 4.2: processing pipeline]" — the last empty figure slot in Chapter 4.
966–968	§5.1	NOTE asking for the V4L2 values. They are: <code>/dev/video4</code> , 1600 × 1200, YUYV, gain 0, white balance 5000 K, gamma 100, auto exposure and auto white balance disabled, a fixed number of initial frames discarded, and <code>exposure_time_absolute</code> bracketed over 1, 5, 20, 50, 100 and 300. State which you settled on.
1230, 1245–1272	§6.2	The long NEW note block. Everything in it is answered by the two figures in Section B above.
1325–1340	§7.1	NEW note block, the two limitations. Figure 7.1 is the evidence for the first.
1354–1358	§7.2	NOTE on October. Figure 7.2 answers it.
1378–1381	§7.3	NOTE on rain. No figure needed: water on the film and droplets in the air both scatter through many events, and multiple scattering destroys the linear polarisation that single Rayleigh scattering produces, so rain removes the quantity being measured rather than merely dimming it.

Claims the figures contradict

1. Line 731, §3.3. "The iso-AoP contours therefore form concentric rings around the solar position." Rings around the sun are contours of constant *scattering angle*, and therefore of constant *degree* of polarisation. The angle of polarisation is perpendicular to the great circle joining the point to the sun and its contours are not rings. Your own Figure 3.2 caption gets this right, so the text contradicts the figure. Line 751 repeats the error.

2. Lines 813–814, §3.3, and lines 1224–1226, §6.2. "the band of low transmission is placed so that it points toward the sun". Measurement says otherwise: binning the rendered image into azimuth wedges puts the darkest wedge at the **antisolar** azimuth every time (sun 0 to dark 180, 90 to 270, 135 to 315, 225 to 45), across elevations of 10, 30 and 50 degrees. Mean intensity toward the sun is 0.489 against 0.293 toward the antisolar point.

The correct statement: the radial arrangement aligns the dark axis with the solar meridian, because there the sky e-vector is tangential while the analyser axis is radial. Both ends darken, and the antisolar end darkens more because the degree of polarisation is low near the sun. The brightness asymmetry between the two ends is what identifies which end holds the sun, and it is why the ambiguity returns once cloud flattens the pattern.

3. Figure 6.3 caption, line 1240, and the List of Figures at line 187. "Because ordinary photographs record no polarisation information" is true only of *unfiltered* photographs, and as written it reads as though the instrument cannot see its own signal. Replace with: "Because a photograph taken without a polarising element records no polarisation information, labelled frames for development and testing are produced in this manner from measured solar geometry rather than drawn from an archive of ordinary sky imagery." The same qualifier is needed at line 1305 in §7.1 and at line 1109 in §5.4. The two must match, body and List of Figures.

4. Figure 6.1 caption mixes two populations. The caption says azimuth spans "approximately 35 to 325 degrees" and the body quotes 5972 daylight positions. Those are different sets: elevation ≥ 0 gives 6740 positions over 35 to 325 degrees, elevation ≥ 5 , which is what the estimator uses, gives 5972 over 50 to 311 degrees. State both and explain the 5 degree floor. Separately, the window 14 May to 14 July is called "the two months leading up to the capture campaign" but the campaign ran 26 to 30 July.

5. Table 6.1 is still the old $n = 60$ table with four rows. The full six-condition matrix has been run at $n = 200$ and is in `outputs/n200_*.txt`. Median absolute error, in degrees: flat clear 0.075, flat thin 0.777, flat heavy 4.103, radial clear 0.075, radial thin 2.120, radial heavy 20.783. Gross failures over 45 degrees, out of 200: 0, 0, 15, 0, 3, 75. **Report the median as the primary statistic** — at $n = 60$ the radial thin cloud mean was 11.59 degrees, at $n = 200$ it is 5.69, while the median held at 2.12 in both.

The result that has to be discussed rather than buried: under a clear sky the two optics are indistinguishable, but **under cloud the radial dome is consistently worse**, with 75 gross failures against 15. The cause is measurable — the radial pattern's contrast is 0.79 against the flat filter's 1.18, so it drowns in clutter sooner. Reporting that plainly is far stronger than leaving it out, and it qualifies the design argument rather than destroying it.

G. Order of work

#	Task	Time	Why now
1	Take the dome photograph and the bare-lens control pair; record the compass heading and the exposure	30 min	Weather-dependent and only you can do it. Everything in Section C waits on it.
2	Copy the three dome photographs into <code>Doc images/</code>	5 min	They exist only in your document right now
3	Write §6.4 around the photograph; insert Figure 6.8	60 min	Fixes the broken §7.1 cross-reference and closes the largest hole
4	Replace Table 6.1 with the $n = 200$ matrix; insert the two §6.3 figures	40 min	The compute is already done
5	Correct the band claim in §3.3 and §6.2; re-insert <code>polarizer_band_demo.png</code> ; insert the wedge profile	40 min	The draft contradicts its own figure
6	Fix the iso-AoP sentence and the "ordinary photographs" qualifier in all four places	20 min	Cheap, and one of them is what prompted the supervisor's question
7	Insert the remaining eight figures and clear the five NOTE blocks	75 min	Removes every visible placeholder
8	Resolve the 4.2 / 4.5 duplicate; renumber Chapters 3 to 7 once	30 min	Do it last, after every figure is placed
9	Rewrite the §4.2 axis paragraph around the three observations	20 min	You have the photographs and the current sentence understates the work
10	Fill Table 2.1 row; fix the citations ([5]→[13], [2]→[10], [3]→[11], [6]→[14], merge [7] and [8])	25 min	Cheap and highly visible

Items 1 and 3 change an examiner's impression more than anything else on the list. Item 8 is the first thing to postpone if time runs short, and the optional Figure 6.7 is the first thing to drop.